

Multi-hop Link Prediction (Ablation)	Multi-hop Link Prediction (KG-LLM)	Multi-hop Relation Prediction (Ablation)	Multi-hop Relation Prediction (KG-LLM)
PROMPT ### Input: Node [node id1] has relation [relation id] with node [node id2]. Node [node id2] has relation [relation id] with node [node id3]. [...] Is node [node id1] connected with node [last node id]? Expected Output ### Response: [Yes / No]	PROMPT ### Instruction: Below is the detail of a knowledge graph path. Is node [node id1] connected with node [last node id]? Answer the question by reasoning step-by-step. Choose from the given options: 1. Yes 2. No ### Input: Node [node id1] has relation [relation id] with node [node id2]. Node [node id2] has relation [relation id] with node [node id3]. [...] Expected Output ### Response: Node [node id1] has relation [relation id] with node [node id2] means [node text1] [relation text] [node text2]. [...] So [node text1] [relation text] [last node text]. The answer is yes.	PROMPT ### Input: Node [node id1] has relation [relation id] with node [node id2]. Node [node id2] has relation [relation id] with node [node id3]. [...] What is the relation between [node id1] and [last node id]? Expected Output ### Response: [relation id]	PROMPT ### Instruction: Below is the detail of a knowledge graph path. What is the relation between [node id1] and [last node id]? Answer the question by reasoning step-by-step. Choose from the given options: 1. [relation text1] 2. [relation text2] [...] ### Input: Node [node id1] has relation [relation id] with node [node id2]. Node [node id2] has relation [relation id] with node [node id3]. [...] Expected Output ### Response: Node [node id1] has relation [relation id] with node [node id2] means [node text1] [relation text] [node text2]. [...] So [node text1] [relation text] [last node text]. The answer is [relation text].